

Preparación PAES

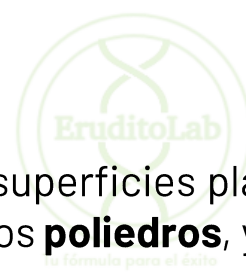
Clase 16: Cuerpos geométricos



Tu fórmula para el éxito

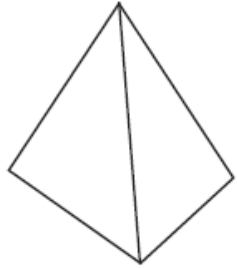
Eje Geometría

Cuerpos geométricos

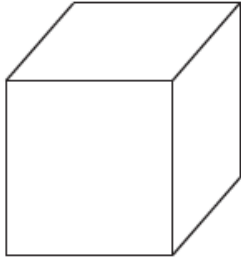


Los cuerpos son figuras de tres dimensiones limitadas por superficies planas o curvas. Dentro de las familias de cuerpos encontramos dos tipos: los cuerpos de caras planas llamados **poliedros**, y los de caras curvas.

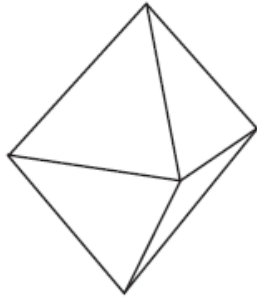
Tetraedro



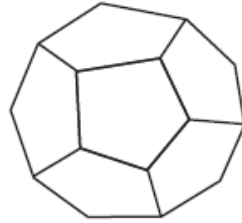
Hexaedro



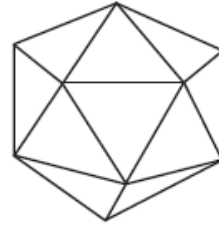
Octaedro



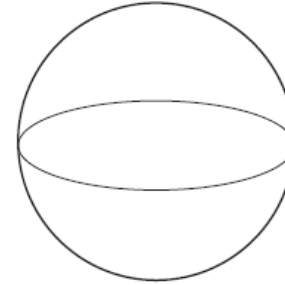
Dodecaedro



Icosaedro



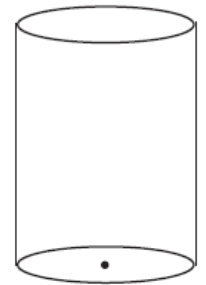
Esfera



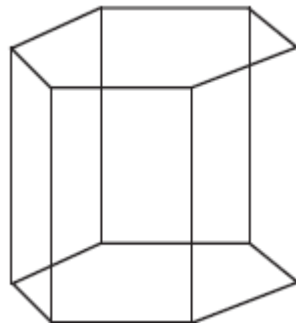
Cono



Cilindro



Prisma.



Eje Geometría

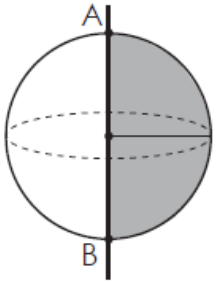
Cuerpos geométricos

- Cuerpos de revolución:
Los cuerpos de revolución se obtienen haciendo girar una superficie plana alrededor de un eje.

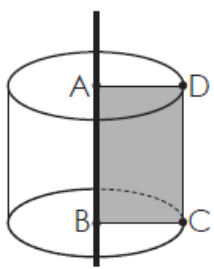


- Cuerpos de traslación:
Se generan por traslación de una superficie plana, en sentido perpendicular al plano que la contiene.

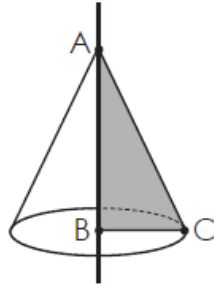
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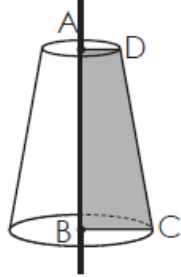
Cilindro



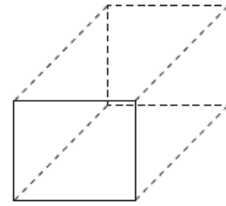
Cono



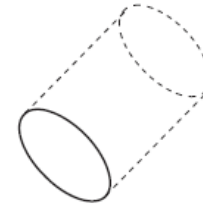
Semi-Cono



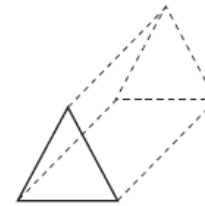
Paralelepípedo



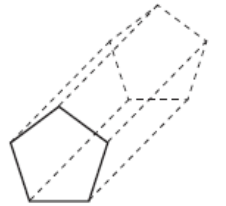
Cilindro



Prisma triangular



Prisma pentagonal

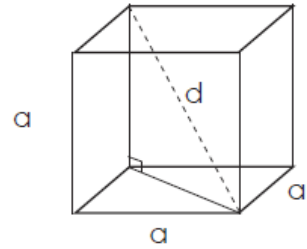


Eje Geometría

Cuerpos geométricos



Cubo

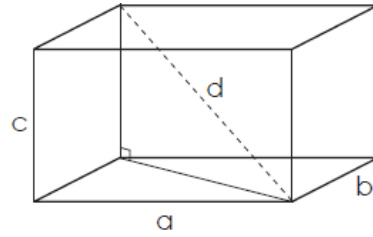


Área: $6 \cdot a^2$

Volumen: a^3

Diagonal (d): $a \cdot \sqrt{3}$

Paralelepípedo

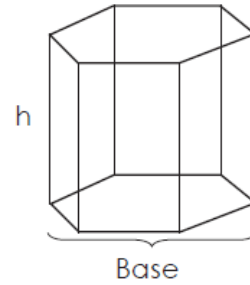


Área: $2(a \cdot b) + 2(b \cdot c) + 2(a \cdot c)$

Volumen: $(a \cdot b \cdot c)$

Diagonal (d): $\sqrt{a^2 + b^2 + c^2}$

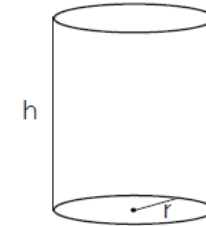
Prisma



Área: Suma de las áreas laterales y basales

Volumen: Área basal $\cdot h$

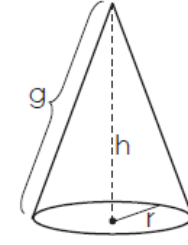
Cilindro



Área: $2 \cdot \pi \cdot r^2 + 2 \cdot \pi \cdot r \cdot h$

Volumen: $\pi \cdot r^2 \cdot h$

Cono

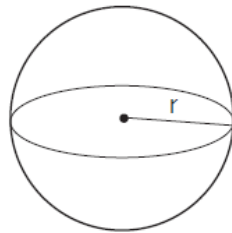


Generatriz: $g = \sqrt{(r)^2 + (h)^2}$

Área: $\pi \cdot r \cdot g + \pi \cdot r^2$

Volumen: $\frac{1}{3} \cdot \pi \cdot r^2 \cdot h$

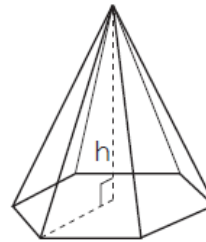
Esfera



Área: $4 \cdot \pi \cdot r^2$

Volumen: $\frac{4}{3} \cdot \pi \cdot r^3$

Pirámide



Área: Suma de las áreas basal y laterales

Volumen: $\frac{1}{3} \cdot \text{Área basal} \cdot h$